Induction Motor

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Induction Motor:-

is an alternating current motor that converts electric energy into mechanical energy. These are the most commonly used motors. These also known as asynchronous motors because they always run slower than synchronous speed.

There are two types of motor. 1. Single Phase motor (1φ
2. Three Phase motor (3φ

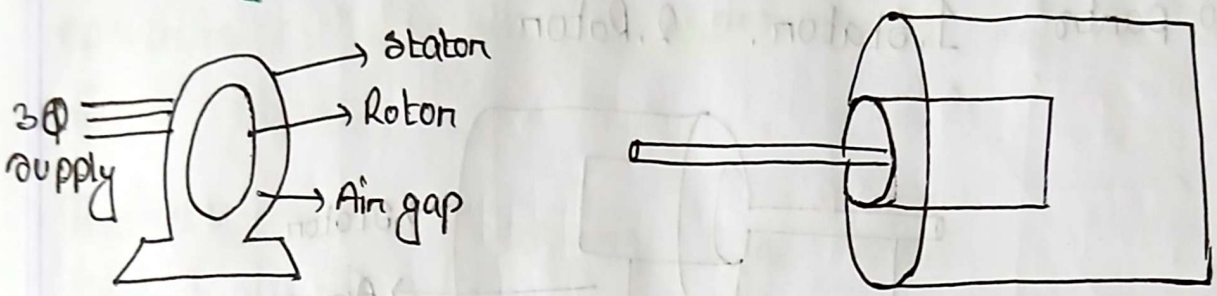
Advantages:-

→ It has simple and rugged construction
→ Relatively low cost.
→ Requires little maintainance
→ It has high efficiency.
→ It has self starting torque.

Disadvantages:-

→ It is a constant speed motor.
→ It's speed can not be changed easily.
→ It's starting torque is inferior to be shunt motor.

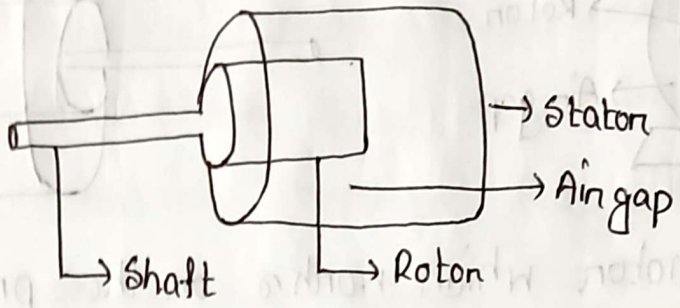
Working Principle of 3phase Induction motor:-



1. The motor which works on the principle of electromagnetic induction is known as the induction motor.
2. A three phase induction motor is based on the production of a rotating magnetic field (r.m.f)
3. When the three phase supply is given to the stator, the rotating magnetic field produced on it.
4. The conductors of the rotor are stationary. The stationary conductor cut the rotating magnetic field of the stator and because of the electromagnetic induction, the EMF induce in the rotor.
5. The magnetic field of stator and rotor will interact to 3Φ induction motor.

Construction of Induction Motor:-

Two parts 1. Stator. 2. Rotor



In stator, Rotating magnetic field is created by continous Northpole/ southpole. Because of this, induced magnetic field is created in Rotor which rotates the motor. There is an air gap, ~~the~~ between stator & rotor. Because of the air gap, ~~the~~ amount ~~of~~ magnetic ~~field~~ stator doesn't pass the same amount of magnetic field to rotor. That's ~~the~~ why there is a speed difference between stator and rotor.

Stator:-

consists of a steel frame which encloses a ~~to~~ hollow cylindrical core made up of thin laminations of silicon steel to reduce hysteresis and eddy current losses.

→ It is the stationary component found in electric motors & generators.

Rotor:-

consists of a stack of laminations in the ~~forme~~ form of a cylinder.

Two types: 1. Squirrel cage rotor
2. Wound rotor

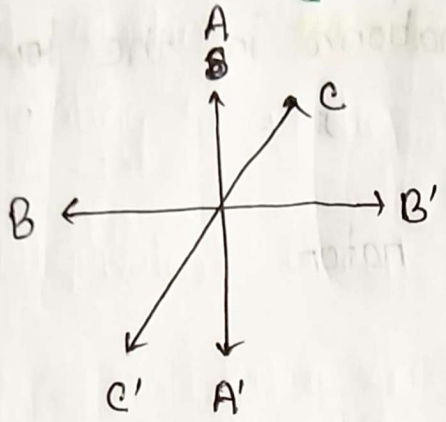
1. Sinusoidal cage Rotor:-

This type of rotor consists of a laminated cylindrical core having parallel slants on its outer periphery.

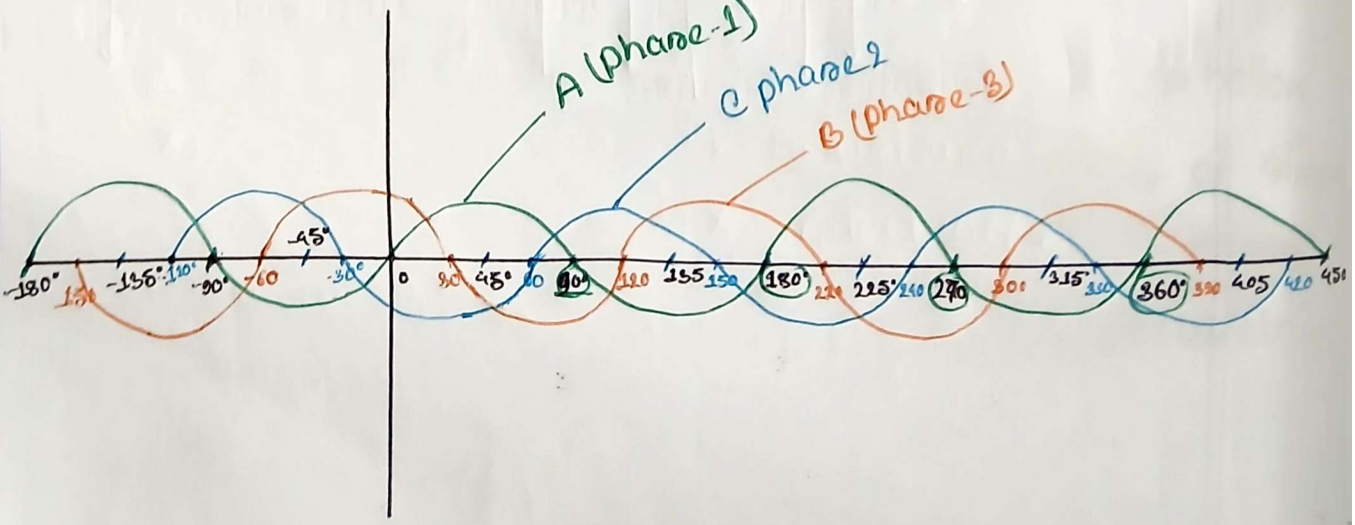
2. Wound Rotor:-

This type of consists of laminated cylindrical core the carries 3 phase winding.

How rotating magnetic field produced in stator:-



A rotating magnetic field is produced in a three phase induction motor by the 3 phase AC currents that flows through the stator windings. The current in each phase is 120° out of the phase with others, create a magnetic field that rotates around the stator at a constant speed. Each alternating ~~pen~~ phase current produces its own flux which is sinusoidal. So, all three fluxes are sinusoidal and are separated from each other by 120°.



Condition-1

In fig-1, A in upward means (direction of phase A in in (+) side but B & C in in negative)

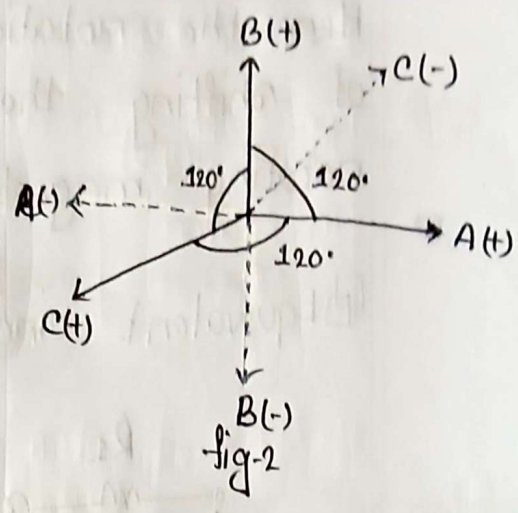
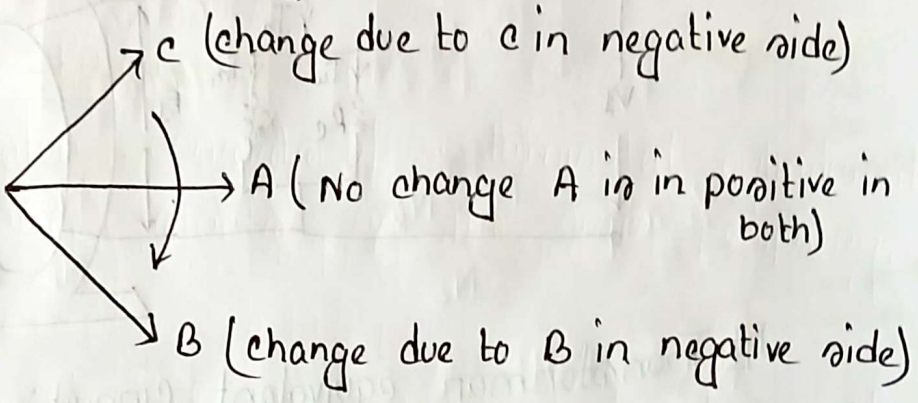


fig-2

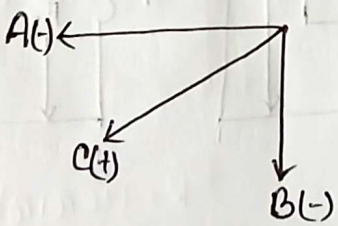
Now compare with fig-2

Rearranging the ~~direction~~ diagram-



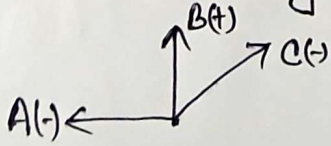
Condition-2

In fig-1 C in in (+) side, but A and B in in negative side. Now comparing with fig-2 and rearranging the diagram



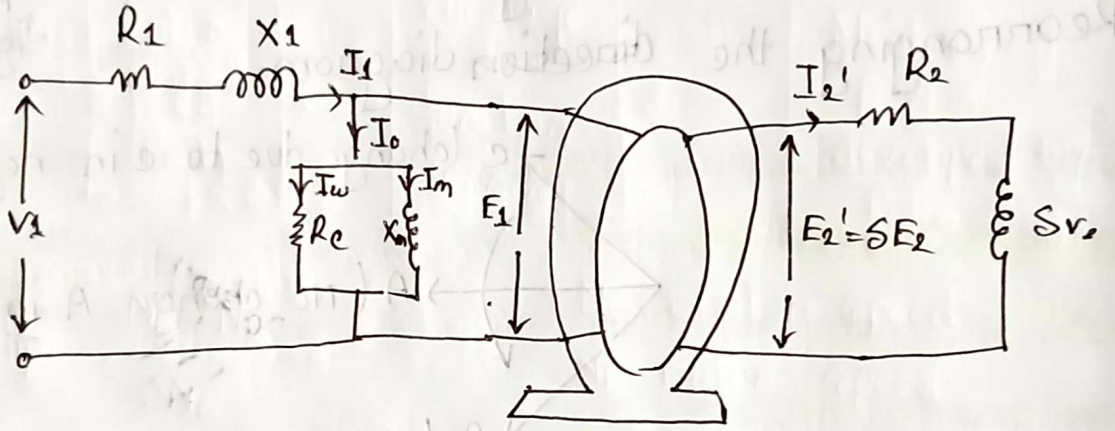
Condition-3

In fig-1 B in in (+) side but A & C in in the negative side, Now compare with fig-2 and rearrange the diagram.

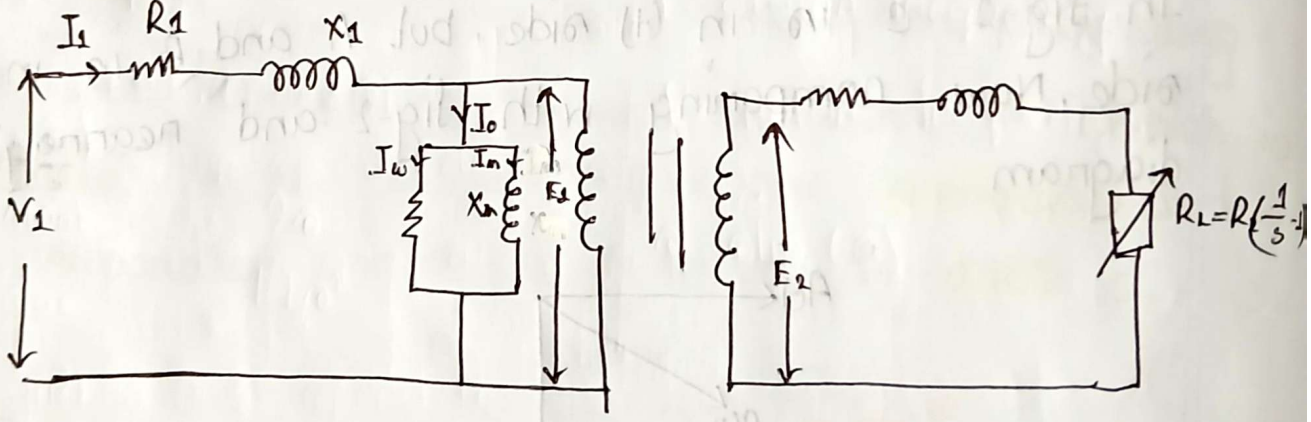


Here, the rotation is continuous. So the above figure of shifting the position of the phase describes how rotating magnetic field is produced.

Equivalent circuit of 3φ IM at any slip:-



Transformer equivalent circuit:-



Slip:

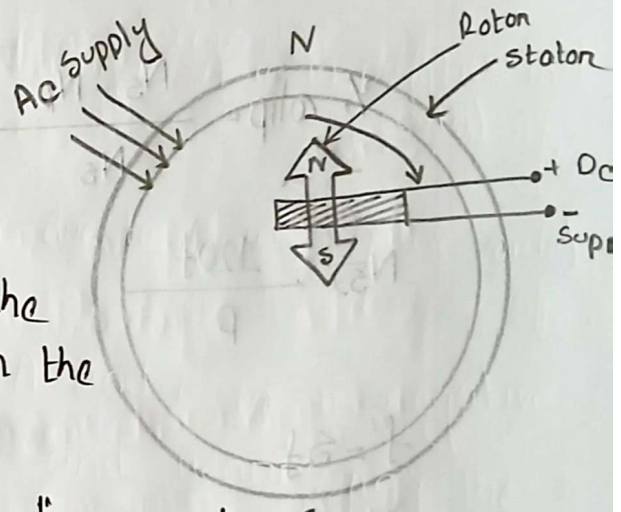
The difference between the synchronous speed N_s and the actual N of the rotor is known as slip.

$$\text{Slip} = \frac{N_s - N_r}{N_s} \times 100$$

[N_s = stator speed
N_r = rotor speed]

Synchronous Motor:-

A synchronous motor is a AC motor where the rotation of the rotor is synchronized with the frequency of the supply current.



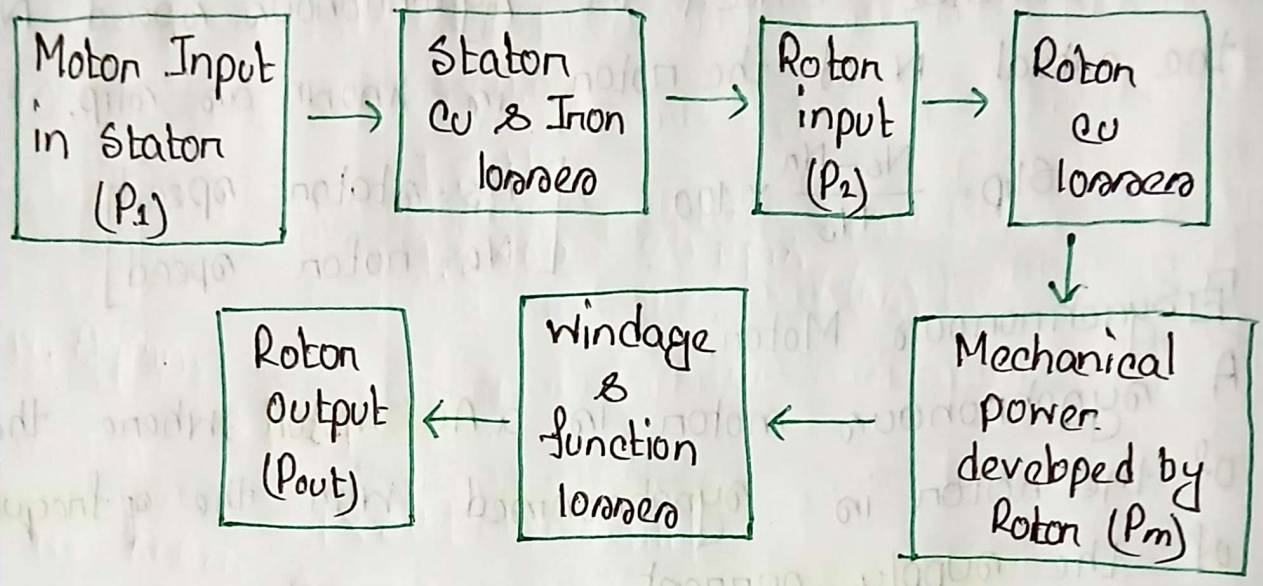
Working Principle:-

Working of synchronous motor depends on the interaction of the magnetic field of the stator with the magnetic field of the rotor.

The stator contains 3 phase winding and supplied with 3 phase power.

1. Stator winding produces a 3 phased rotating magnetic field.
2. DC supply is give to the rotor.
3. The rotor enters into the rotating magnetic field produced by the stator winding and rotates in synchronization.

Power stages in 3 phase Induction Motor:-



$$\% \text{ slip} = \frac{N_S - N_R}{N_S}$$

N_S = Stator speed
N_R = rotor speed

$$N_S = \frac{120f}{P}$$

P = pole
f = Stator frequency

$$f' = sf$$

f' = rotor frequency.

$$\frac{T_{start}}{T_{max}} = \frac{2s}{1+s^2}$$

s = slip (a)

$$s = \frac{R_2}{X_2}$$

R = resistance
X = inductance.

34.3

A 4 pole, 3phase induction motor operates from a supply whose frequency is 50Hz.

i. $N_S = ?$

ii. $N_r = ?$ where slip = 0.04.

iii. the frequency of rotor currents when slip = 0.03

iv. " " " " " when at standstill.

Given,

$P = 4$

$f = 50H.$

i. Stator speed $N_S = \frac{120f}{P}$

$= \frac{120 \times 50}{4}$

$= 1500$ rpm. (Ans).

ii. Rotor speed; N_r

We know, $S = \frac{N_S - N_r}{N_S}$

$\Rightarrow 0.04 = \frac{1500 - N_r}{1500}$

$\Rightarrow 1500 - N_r = 60$

$\Rightarrow N_r = 1440$ rpm (Ans)

iii. We know,

$$f' = sf$$
$$= 50 \times 0.03$$
$$= 1.5 \text{ Hz}$$
$$= 1.5 \times 60 = 90 \text{ rpm} \quad \text{(Ans)}$$

iv. Again,

$$f' = sf$$
$$= 1 \times 50 \text{ [Standstill]}$$
$$= 50 \text{ Hz} \quad \text{(Ans)}$$

34.4

A 3φ IM is wounds for 4 poles and supplied from 50Hz system. calculate

i. Synchronous speed.

ii. the rotor speed, when slip = 4%.

iii. rotor frequency when rotor runs at 600 rpm.

i. Synchronous speed, $N_s = \frac{120f}{p}$

$$= \frac{120 \times 50}{4}$$
$$= 1500 \text{ rpm.} \quad \text{(Ans)}$$

ii. $\%S = \frac{N_s - N_r}{N_s}$

$\Rightarrow N_r = -(\%S \times N_s) + N_s$

$= N_s - (\%S \times N_s)$

$= 1500 - (0.04 \times 1500)$

$= 1440$ rpm. (Ans)

iii. Given,

$N_r = 600$ rpm.

$N_s = 150$ rpm.

$\therefore S = \frac{N_s - N_r}{N_s}$

$= \frac{1500 - 600}{1500}$

$= 0.6$

$\therefore$ Rotor frequency, $f' = Sf$

$= 0.6 \times 50$

$= 30$ Hz. (Ans)

34.14(a)

A 3φ, 400/200V Y-Y connected wound-rotor induction motor has 0.06Ω rotor resistance & 0.3Ω standstill reactance per phase. Find the additional resistance required in the rotor circuit, to make the starting torque equal to the maximum torque of the motor.

We know,

$$\frac{T_{start}}{T_{max}} = \frac{2a}{1+a^2}$$

$$\Rightarrow 1 = \frac{2a}{1+a^2} \quad [\text{standstill}]$$

$$\Rightarrow (a-1)^2 = 0$$

$$\Rightarrow a = 1.$$

Again,

$$a = \frac{R_2 + r}{X_2} \quad [r = \text{additional resistance}]$$

$$\Rightarrow 1 = \frac{0.06 + r}{0.3}$$

$$\Rightarrow 0.06 + r = 0.3$$

$$\Rightarrow r = 0.3 - 0.06$$

$$= 0.24\Omega \text{ (Ans)}$$

34.27

The power input to the motor of 440v, 50Hz, 6 pole, 3 phase, induction motor is 80kw. The motor electromotive force is observed to make 100 complete alternations per minute, calculate

i. the slip
ii. the motor speed.
iii. motor cu loosses per phase.

Given,

$f = 50Hz$

$P = 6$ pole.

the frequency of the motor $\frac{100}{60}$

$= 1.67 Hz.$

∴ We know, $f = Sf'$

$\Rightarrow S = \frac{f'}{f'}$

$= \frac{50}{1.67} = \frac{1.67}{50}$

$= 30 \quad = 0.03.$

Here, $N_S = \frac{120f}{P}$

$= \frac{120 \times 50}{6} = 1000 rpm$

Again,

$$S = \frac{N_s - N_r}{N_s}$$

$$\Rightarrow 0.03 = \frac{1000 - N_r}{1000}$$

$$\Rightarrow 1000 - N_r = 30$$

$$\Rightarrow N_r = 1000 - 30$$

$$= 970 \text{ rpm.}$$

$$N = N_s(1-S)$$

$$= 1000(1-0.3)$$

$$= 967 \text{ rpm}$$

Rotor cu loss per phase = (S x rotor input)

total rotor power input = 80kw.

∴ rotor power input per phase = $\frac{80}{3} = 26.67$ kw.

∴ Cu loss per phase = (0.03 x 26.67)

~~= 0.2667 / 0.293kw~~

= 0.8001. (Ans)

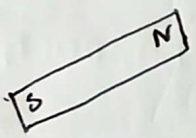
Synchronous vs Induction motor:-

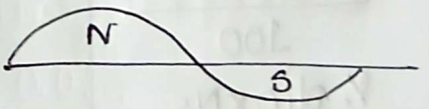
Synchronous	Induction
1. Speed remain constant.	1. Speed decreases with load.
2. Power factor can be made to operate from lagging to leading	2. Operates a lagging power factor.
3. Economical speed below 300 rpm.	3. Economical speed above 600 rp
4. No self starting torque.	4. Self starting torque.
5. Construction is complicating.	5. Construction is simple.
6. It's torque is more sensitive	6. It's torque is less sensitive
7. Can be used as synchronous condenser.	7. Can't be used as synchro nous condenser.

Synchronous Speed:

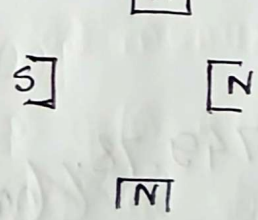
The speed at which the rotating magnetic field in stator revolves is called synchronous speed (N_s)

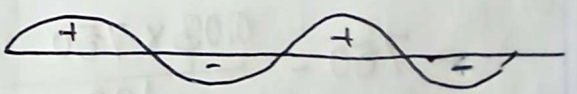
2 pole →





4 pole →





34.5

A 12 pole, 3 phase alternator driven at a speed of 500 supplies power to an 8 pole, 3 phase induction m If the slip of the motor, at full load is 3%, calcula the full load speed of the motor.

Let,

N = Actual motor speed.

$\therefore$ Supply frequency, $f = \frac{NP}{120}$

$= \frac{500 \times 12}{120}$

$= 50 Hz$

$\therefore N_s = \frac{120 \times 50}{P}$

$= \frac{120 \times 50}{8}$

$= 750 rpm.$

$\therefore \%\ slip = \frac{N_s - N}{N_s} \times 100$

$\Rightarrow N_s - N = \frac{\%\ slip \times N_s}{100}$

$\Rightarrow N = N_s - \frac{\%\ slip \times N_s}{100}$

$= 750 - \frac{0.03 \times 750}{100} = 749.78.$ (Ans)

34.19

An 8 pole, 50Hz, 3phase slip ring induction motor has effective motor resistance of 0.08Ω. Stalling speed is 650rpm. How much resistance must be inserted in the rotor phase to obtain the maximum torque at starting?

We know,

$$N_s = \frac{120f}{P}$$

$$= \frac{120 \times 50}{8}$$

$$= 750 \text{ rpm.}$$

$$\text{Slip, } s = \frac{750 - 650}{750}$$

$$= 0.13 = 13.33\%$$

Now,

$$s = \frac{R_2}{X_2}$$

$$\therefore X_2 = s \times R_2$$

$$= 0.13 \times 0.08$$

$$= 0.6\Omega.$$

Again,

$$\frac{T_{start}}{T_{max}} = \frac{2a}{1+a^2}$$

$$\Rightarrow 1 = \frac{2a}{1+a^2} \quad [T_{start} = T_{max}]$$

$$\Rightarrow (a-1)^2 = 0$$

$$\Rightarrow a = 1$$

let, r be the external resistance

$$a = \frac{R_2 + r}{X_2}$$

$$\Rightarrow 1 = \frac{0.08 + r}{0.6}$$

$$\Rightarrow 0.08 + r = 0.6$$

$$\Rightarrow r = 0.6 - 0.08$$

$$= 0.52\Omega \text{ per phase.}$$

(Ans)

Chapter-5

Aneela Ahmed (C233491) Stepper motor

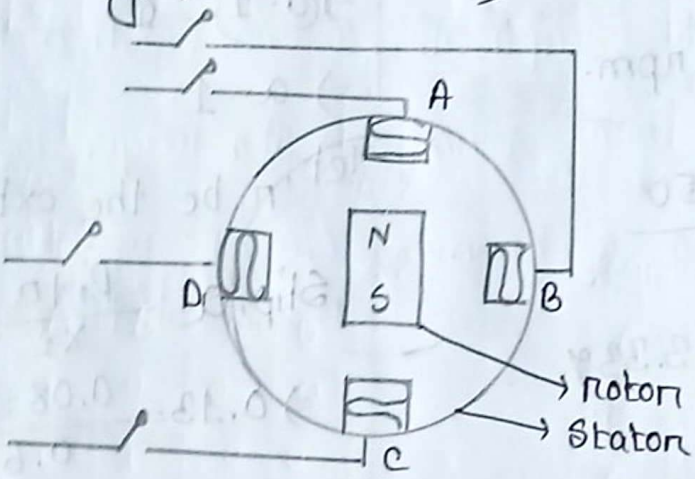
Stepper motor:-

A stepper motor also known as stepper motor or stepping motor, is an electrical motor that rotates in a series of small angular steps instead of continously.

Stepper motor has two parts

- Stator
- rotor (permanent magnet is used)

→ A steppe motor is an electric motor and have a stationary part (stator) and a moving part (rotor).



Here every pole aren't energited together. So, it's easier to control the step. A stepper motor usually can revolve 200 steps to complete a cycle. So, each step $\frac{360^\circ}{200} = 1.8^\circ$ step angle.

Stepper mode can have maximum 90° angle. Minimum step angle is 0.72° and maximum step angle is 90°.

1.8°, 2.5°, 7.5°, 15° step mode are used.

Stepper motor working Principle:

By energizing one or more of the stator phases, a magnetic field is generated by the current flowing in the coil and the rotor aligns with this field. By supply different phases in sequence, the rotor can be rotated by a specific amount to reach the desired final position. At begining, coil A is energized and the rotor is aligned with the magnetic field it produces. When coil B is energized the rotor rotates clockwise by 60° to align with the new magnetic field. The same happens when coil C is energized.

Stepper motor driving modes:

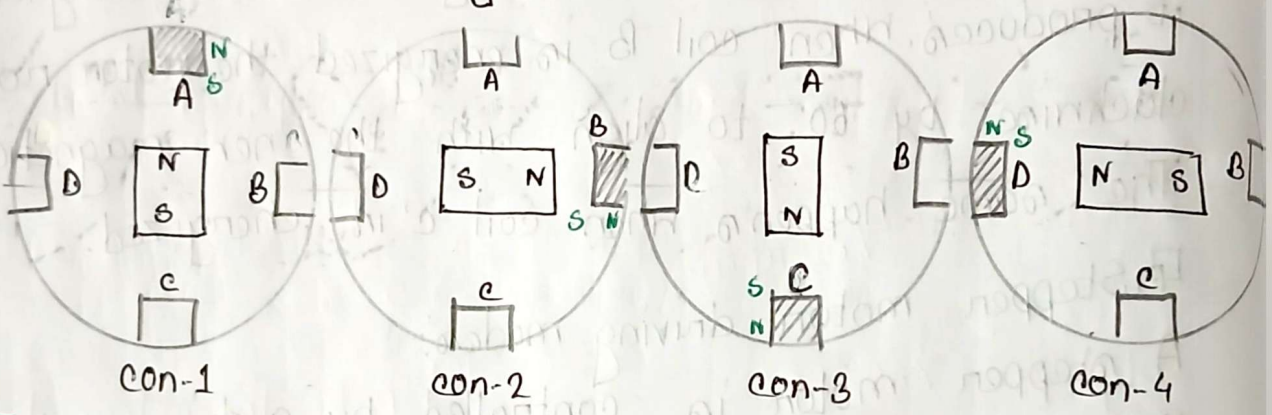
A stepper motor is controlled by a driver which sends the pulse into motor causing it to turn.

Stepper motor driving modes -

1. Single coil excitation mode / wave mode
2. Full step mode
3. Half step mode.
4. Micro step mode.

Single coil excitation / wave mode:-

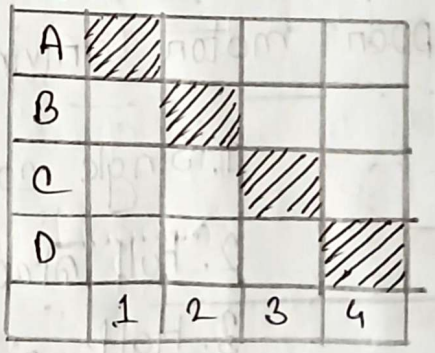
In this mode, only one phase is energized at a time. If coil (pole) A is energized, then north pole (N) and south pole (S) is created here. As a result, the rotor is aligned with the magnetic field generated by the coil. It is the same for coil B, C, D. Here the rotor spins 90° clockwise with the magnetic field generated by the coil.



Truth Table:-

Condition	A	B	C	D
1	1	0	0	0
2	0	1	0	0
3	0	0	1	0
4	0	0	0	1

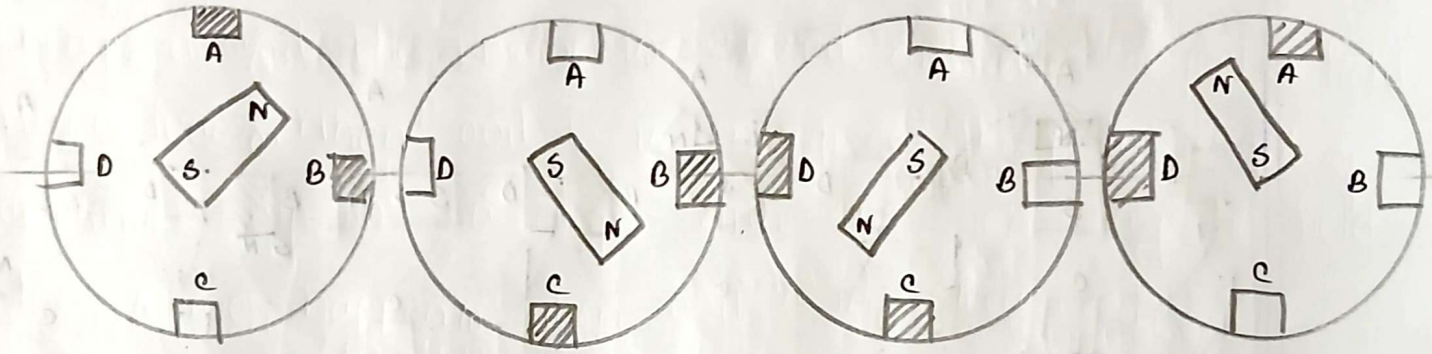
Logic Diagram:-



Full Step mode:

Two phases are always energized at the same time.

The steps are similar to the wave mode ones. The motor is able to produce a higher torque since more current is flowing in the motor and a stronger magnetic field is generated. Step angle is 90°.

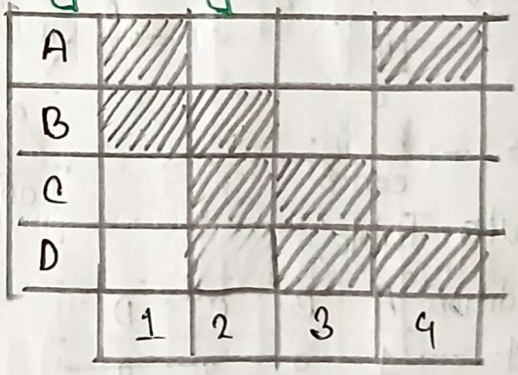


con-1 con-2 con-3 con-4

Truth Table:

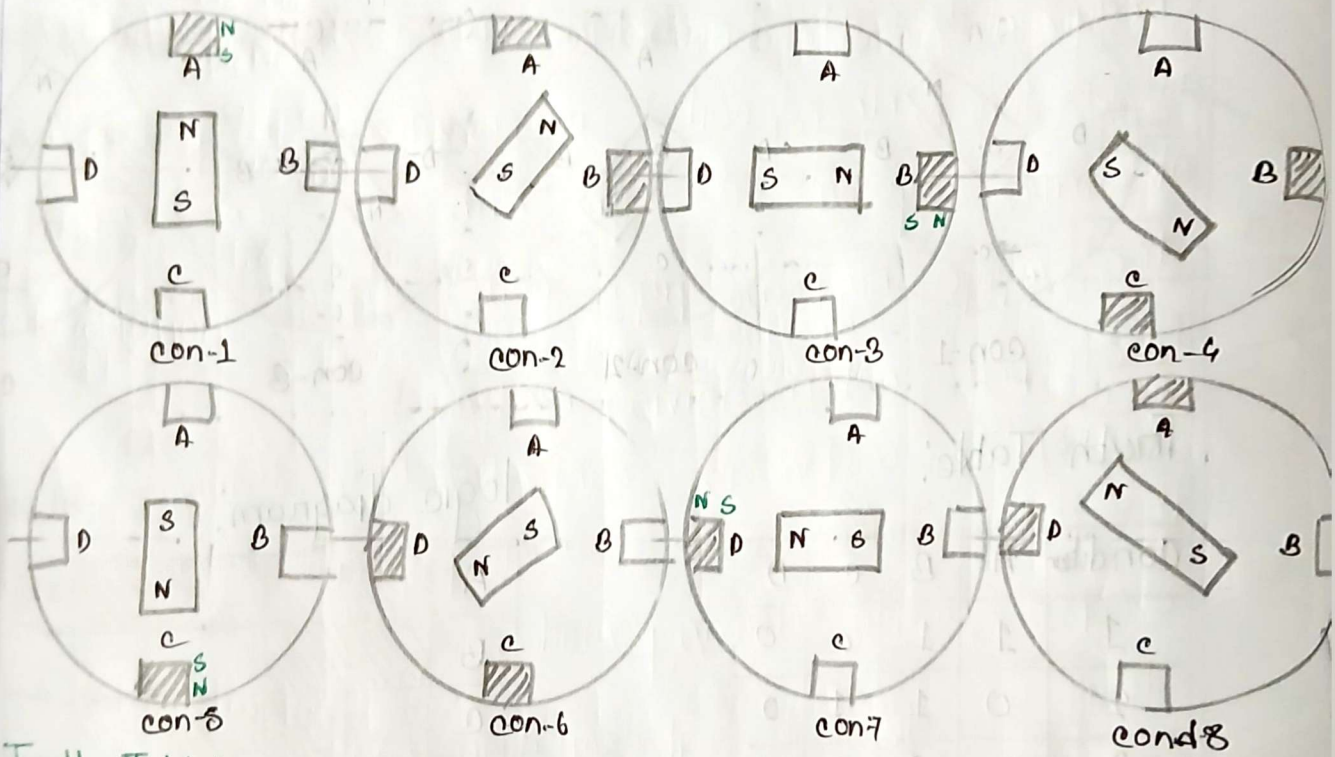
Condition	A	B	C	D
1	1	1	0	0
2	0	1	1	0
3	0	0	1	1
4	1	0	0	1

logic diagram:



Half-Step mode:

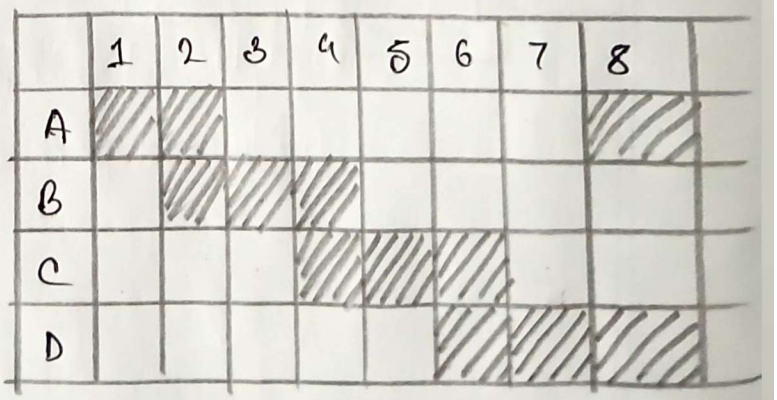
It is the combination of wave and full step mode. The step size is to be reduced by half (in this case 45° instead of 90°). It higher (torque) when both phases are neglected and weaker when only one phase is energized The torque produce the motor is not constant.



Truth Table:-

Condition	A	B	C	D
1	1	0	0	0
2	1	1	0	0
3	0	1	0	0
4	0	1	1	0
5	0	0	1	0
6	0	0	1	1
7	0	0	0	1
8	1	0	0	1

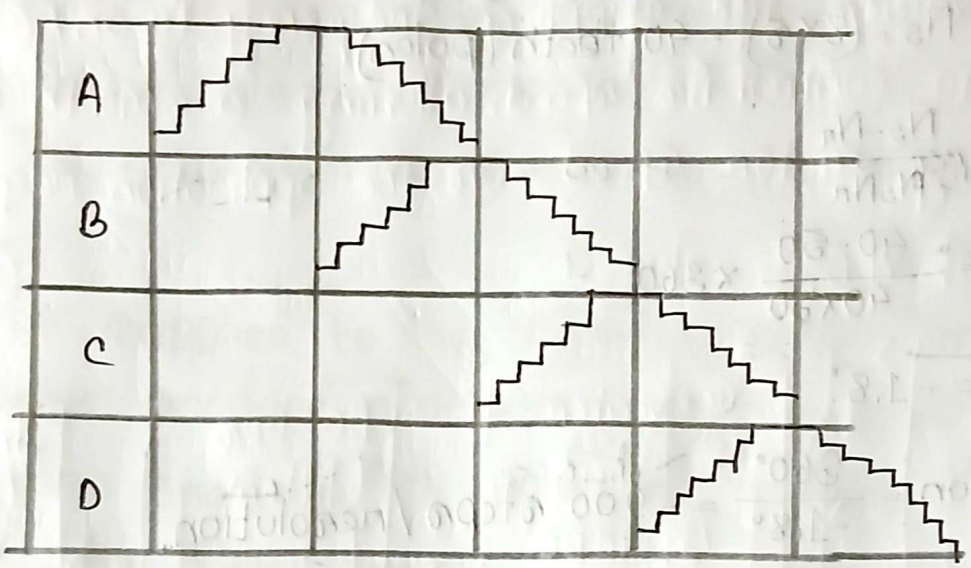
Logic Diagram:-



Micro-stepping mode:

The step angle are very small it almost looks like continous. If powe in stator is supplied slowly so the step change slowly.

logic Diagram:-



Formula:-

$$\text{Step angle } (\beta) = \frac{N_s - N_R}{N_s N_R} \times 360^\circ$$

$$\text{Resolution} = \frac{360^\circ}{\text{step angle}} = \frac{\text{No of steps}}{\text{revolution}}$$

N_S = Stator poles (teeth)

N_R = rotor (or) Poles teeth

β = step angle

A hybrid VR stepping motor has 8 main poles which have been constructed to have 5 teeth each. If the motor has 50 teeth, calculate the stepping angle and resolution.

Here,

$N_r = 50$ teeth (poles)

$N_s = (8 \times 5) = 40$ teeth (poles)

$$\beta = \frac{N_s - N_r}{N_s N_r}$$

$$= \frac{40 - 50}{40 \times 50} \times 360$$

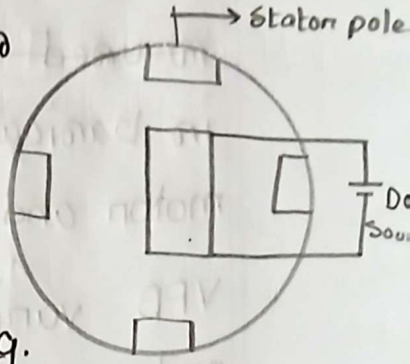
$$= -1.8^\circ$$

$$\text{Resolution} = \frac{360^\circ}{-1.8^\circ} = -200 \text{ steps/resolution.}$$

Working Principle of Synchronous Motor:-

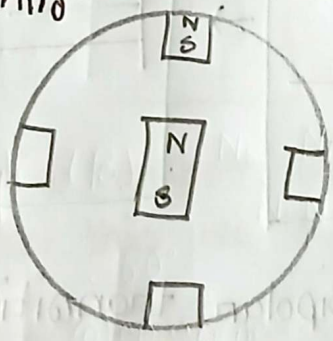
A synchronous motor has two main parts

1. Stator - supplied with AC
2. Rotor - supplied with DC



→ DC power is applied to the rotor winding. This creates a constant magnetic field on the rotor. The rotor behaves like a magnet without its own N-S poles.

→ AC is supplied to the stator winding. Then the stator currents produce a rotating magnetic field (RMF). This RMF also creates N-S poles that rotate at synchronous speed like this



→ When the stator's rotating magnetic field sweeps past the rotor, the rotor's magnetic poles try to align with the stator poles. Once they lock in, the rotor is pulled into synchronism.

→ After locking, the rotor rotates at the same speed as the rotating magnetic field. This speed is called synchronous speed, and the motor continues running at this exact speed - no slip occurs.

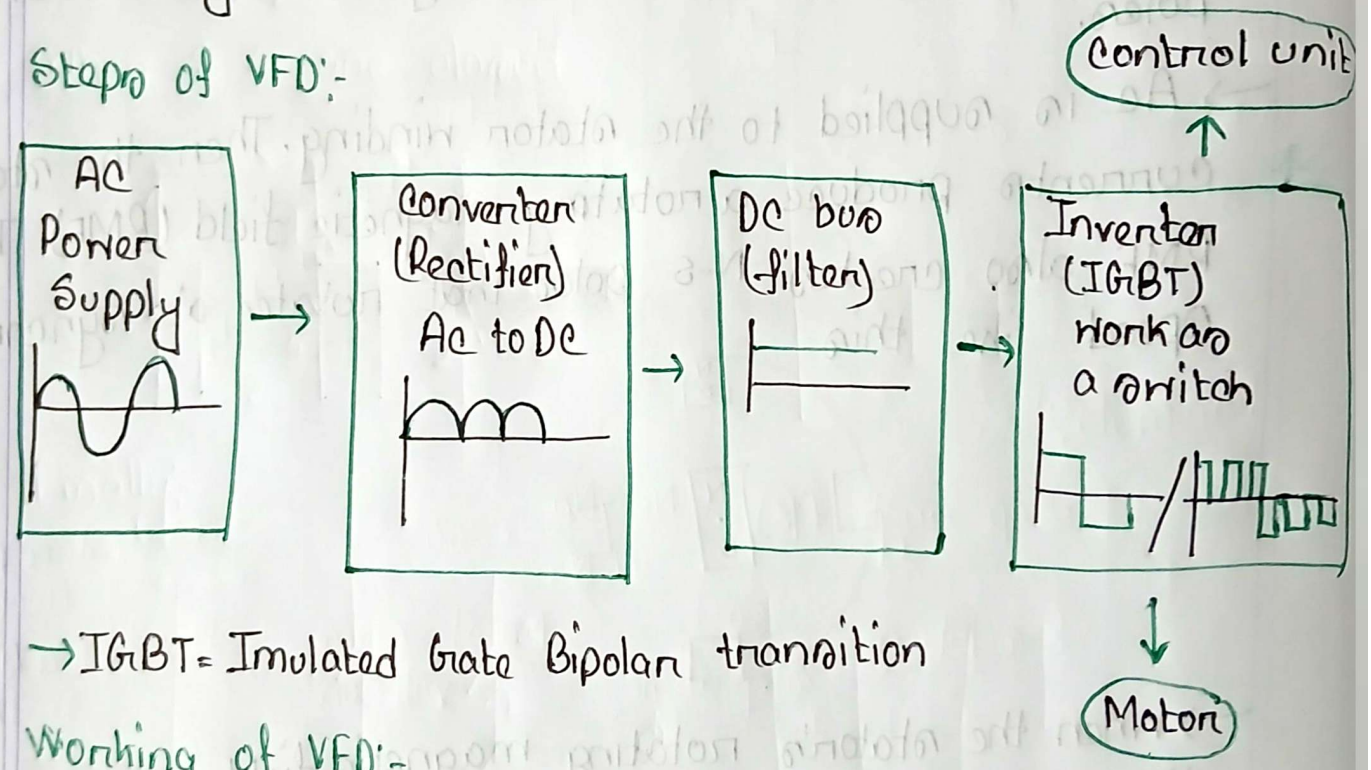
Variable frequency drive:- [VFD]

~~is used for~~

is basically used for controlling the speed of induction motor and synchronous motor.

VFD varies the applied frequency to an AC motor in order to control its speed, allowing a smooth startup and adjusting motor speed as the application required.

Steps of VFD:-

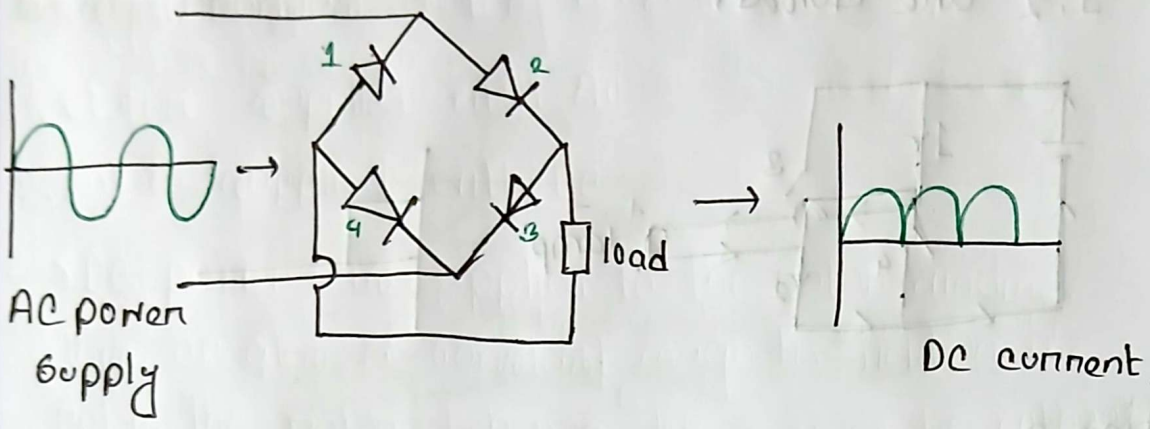


→ IGBT = Insulated Gate Bipolar transition

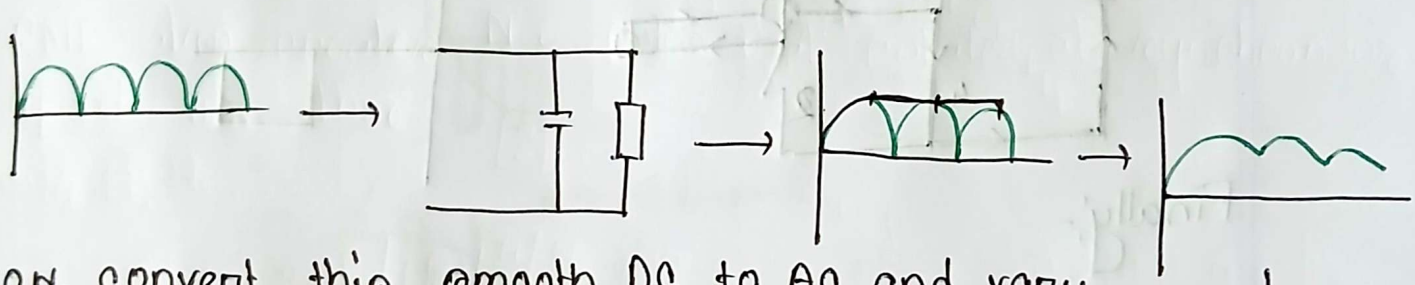
Working of VFD:-

Converter → the rectifier is used to convert AC power supply to DC.

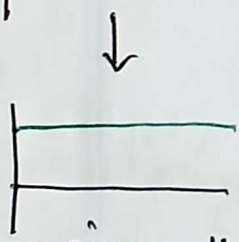
1,2,3,4 → diode



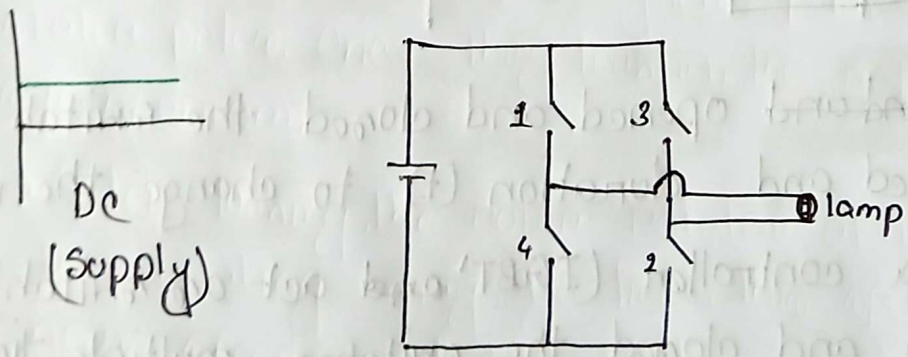
There is ripple in DC (current), so, we need to smooth out those ripples to clean up the DC electricity for these we use a DC bus which is actually a filter.



NOW, convert this smooth DC to AC and vary the frequency for this we need inverter. Inverter are basically IGBT which act as switch that can turn on and off supper fast.



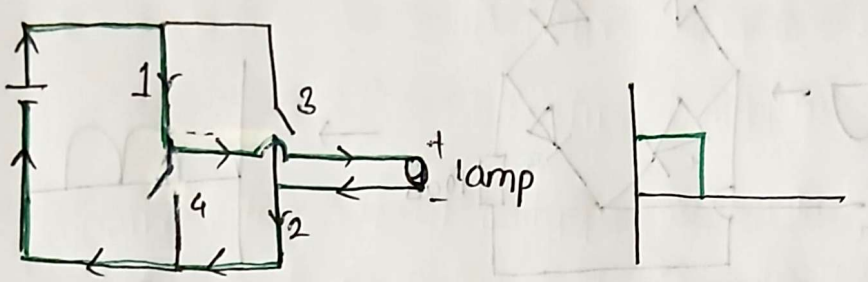
using voltage regulation



1,2,3,4 → IGBT act as a switch

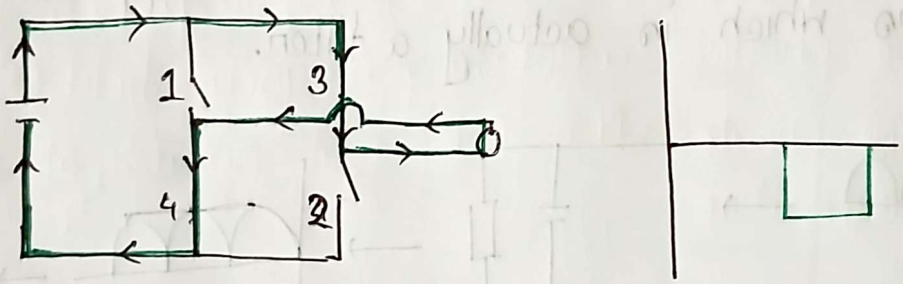
Condition-1

When 1, 2 are closed,

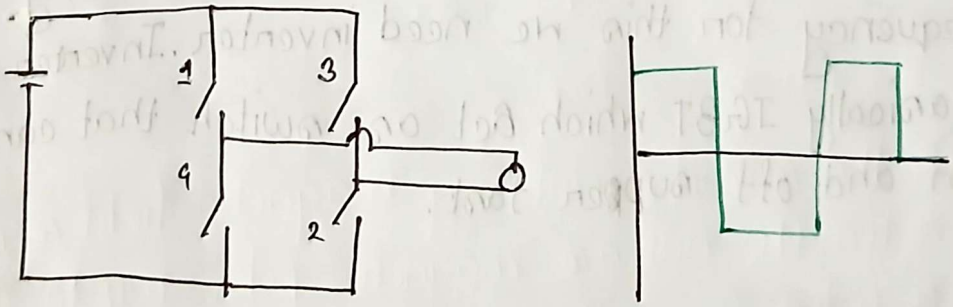


Condition-2

When 3, 4 are closed



Finally:-

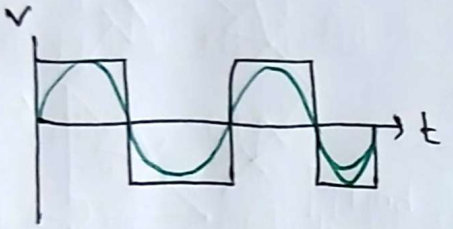


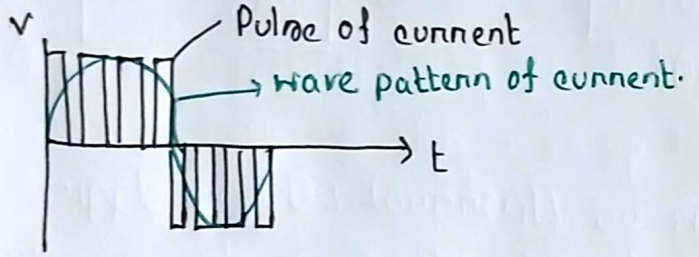
Here, we ~~open and~~ opened and closed the switch at different speed and duration (t) to change the waveform

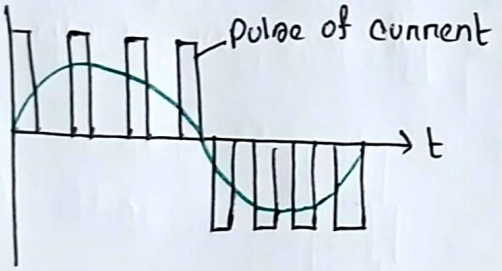
If we need a controller (IGBT ~~and~~ act as switch) to rapidly open and closed the switches multiple times per cycle in a pulsating pattern, each pulse varying

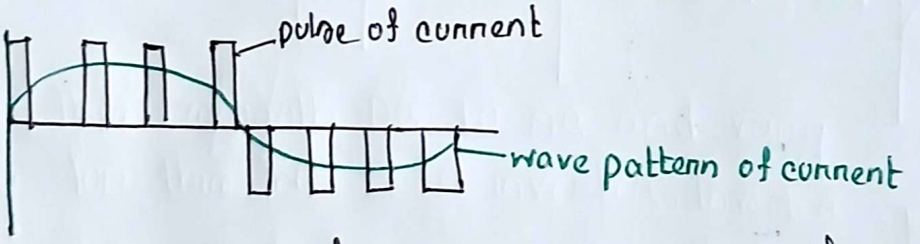
in width. This is known as PWM (Pulse width modulation).

So, the output is









Here we can control the frequency by controlling the timing at the switch.

If t is increased, frequency is decreased $f = \frac{1}{t}$

If t is decreased, frequency is increased.

If frequency is decreased or increased then speed is also increased or decreased.

$$N_s = \frac{120f}{P} \qquad N_s \propto f$$

So, by controlling frequency, we can control the speed of motor.